ReCAP: CAPTCHA Solving for Native GUI Agents

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Paper version: arXiv 2603.23559v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *CAPTCHA solving remains a major challenge.*

Native agents lack CAPTCHA skills; specialized solvers lack general GUI ability and need many model calls per challenge.

Motivation: three gaps

1. **Native agents** fail: noisy OCR, fine control.
2. **Pipelines** (Halligan, Oedipus): standalone, many calls.
3. **No closed-loop data:** CAPTCHA resources are eval-only.

Evaluation setup

Dynamic CAPTCHA · 1,000 | 7 challenge types | 5-step budget (8 grid) | Real-world · 26 types | Qwen3-VL 8B/32B base | Qwen2.5-VL-72B expert | 1 epoch SFT | 230k train samples

- **Online** interactive SR; offline GUI results off-chart.
- **Held out:** real-world set untrained; Halligan built for it.

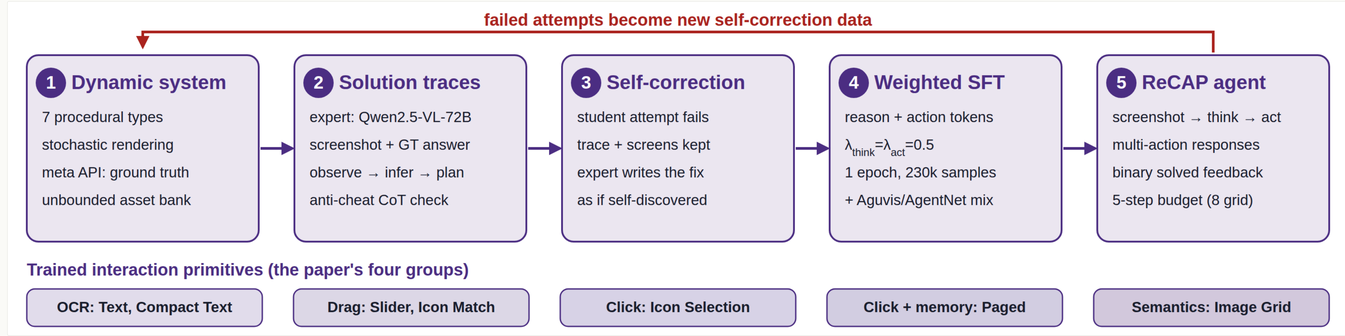
Our read (CUA Radar)

64	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out env; strong baselines; cost; ablations; code announced. **Not reported:** seeds/variance; full budget match.
- **Flags:** Halligan budget partial; CoT is answer-conditioned supervision.
- Pattern C21 (reasoning SFT). Authors' numbers, not rerun.

Method: ground-truth-conditioned reasoning-action data, then weighted SFT

- **Sees / acts:** screenshot in; click, drag, type out — several actions per response.
- **Data:** expert (Qwen2.5-VL-72B) writes CoT with the ground truth given; checked so the CoT does not reveal the answer.
- **Signal:** failed student attempts become retrospective self-correction traces; unified loss balances thinking vs action tokens.



Our redraw (not the paper's figure): procedural environment → two trace generators → weighted SFT → the deployed agent loop.

$$\mathcal{L}=-\frac{\lambda_{\mathrm{think}}}{|T|}\sum_{t\in T}\log p_\theta(y_t\mid x,y_{

Unified loss over reasoning tokens T and action tokens A; λthink = λact = 0.5

Key results: held-out dynamic CAPTCHA, online SR (%)

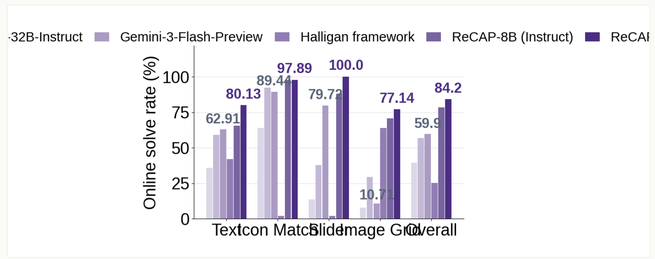
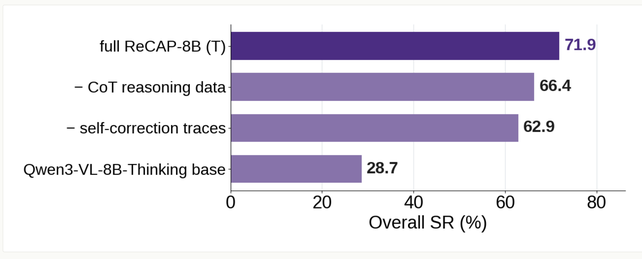


Table 1, redrawn: 1,000 held-out instances, online solve rate; no variance reported.

Overall 39.30 → 84.20 over the Qwen3-VL-32B-Instruct base

SO WHAT → *Real-world (zero-shot, 26 types): +14.50 pp (8B Instruct) and +11.42 pp (32B Instruct) vs bases.*

Ablations: ReCAP-8B (Thinking), overall SR (%)



Tables 4–5, redrawn: same base, same data, one component removed; per-type pairs show the largest self-correction gains.

SO WHAT → *CoT data: 66.40 → 71.90. Self-correction: 62.90 → 71.90; steps stay low (1.61 → 1.54 and 1.60 → 1.54). Per the paper, self-correction gains are largest on Text (+16.77pp) and Slider (+21.9pp); CoT helps all seven types. Data scale saturates: 62.90 / 71.90 / 72.20 at 50% / 100% / 150%.*

Headline Numbers

84.20%
held-out CAPTCHA SR
vs 39.30% Qwen3-VL-32B-Instruct base

78.60% 8B Instruct overall

+14.50 pp real-world vs base

1.33 avg steps, 8B Inst

The paper's own controls

CONTROL	RESULT	VS
Held-out test set	1,000 fresh cases	150k train traces
Real-world zero-shot	+14.50 pp (8B Inst)	Halligan tuned to it
Step budget 5/8	1.33 calls (8B Inst)	Halligan ≥3–5 steps
GUI kept (Tbl. 3)	32B: UI-Vision IoU up	8B: ScreenSpot drops

Takeaway

Answer-conditioned CoT and self-correction traces turn Qwen3-VL into a CAPTCHA solver: 84.20 held-out, +14.50 pp real-world; transfer mixed, 32B keeps GUI scores.

150k solution traces | 10k correction traces | 72.20 SR at 150% data

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Paper | Code